

CubeSat Attitude Control

Fall 2025 Final Report

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Abstract

In this project, we designed an attitude control system for the Cooper Union Space Launch Initiative's (CUSLI) CubeSat. CubeSats are small satellites which are often used by universities to provide access to space, and CUSLI is building a CubeSat to test a scientific payload which consists of an e-ink screen for thermal regulation. To properly perform experiments, charge its batteries, and communicate with the Earth, the CubeSat needs to be able to control its orientation in space.

We designed a system to do this using magnetorquers, which are actuators which can be used to produce torques to turn the satellite. We first found optimal designs for magnetorquers which meet all technical constraints and maximize a score function. We then physically assembled magnetorquers to verify our calculations of their properties. Over the course of the next semester, we plan to further develop a 1-axis magnetorquer demonstration in order to fully characterize magnetorquer torques, as well as design and simulate a 3-axis control algorithm for the CubeSat using magnetorquers.

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Introduction

CubeSats are a class of small satellites that use a standard size and form factor [1]. Since the first CubeSat launch in 2003 [2], they have made space accessible to universities and small organizations with limited budgets. CubeSats can typically be divided into a few sub-systems: structure, power, communications, command and data handling, thermal, and attitude determination and control systems (ADCS). The ADCS sub-system is responsible for determining and controlling the orientation of the CubeSat in space. However, commercial off-the-shelf (COTS) ADCS systems typically cost thousands of dollars, so the Cooper Union Satellite Launch Initiative's (CUSLI) CubeSat needs a low-cost alternative to successfully perform their experiment. CUSLI's experimental payload is an E-ink screen which will be tested as a thermal control system for satellites in space. To perform the experiment, the CubeSat needs to be able to detumble after launch, point its solar panels towards the sun, point its antenna towards the earth, and point its payload towards the sun. CUSLI has already designed an attitude determination system, so they want a cost-effective control system that can achieve these requirements while remaining robust and reliable. Other stakeholders in this system include the launch provider requiring strict safety specification, funding sources expecting technical feasibility and value, and future CUSLI team members needing clear documentation.

Background

Actuation

The ADCS landscape for CubeSats has evolved significantly in the past 20 years, yet fundamental constraints remain unchanged. One important actuator, in use historically and currently, is the magnetorquer. Magnetorquers are coils of wire, possibly with a ferromagnetic core, which generate torque according to the equation $\vec{\tau} = \vec{m} \times \vec{B}$, where $\vec{\tau}$ is the torque vector, $\vec{B}$ is the Earth's magnetic field vector, and $\vec{m}$ is the magnetic dipole vector of the magnetorquer, which measures the strength of the magnetorquer. $\vec{m}$ can be calculated as $\vec{m} = nI\vec{A}$, where n is the number of coils, I is the current through the coils, and A is the vector area of the coils.

Figure 1 shows a diagram of a magnetorquer along with its magnetic field [3].

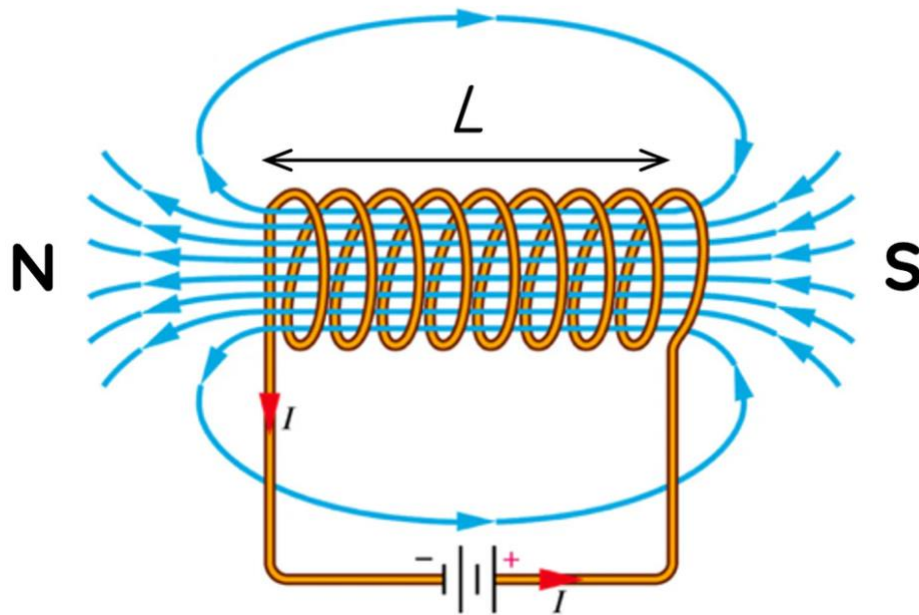


Figure 1: Diagram of Solenoid and its Magnetic Field [3]

One fundamental limitation of magnetorquers is that at a given time, they can only produce torques orthogonal to the Earth's magnetic field vector, giving them limited control over all three axes. In addition, the torques they produce are typically thousands of times smaller than those produced by other possible actuators such as reaction wheels or thrusters. Despite this limitation, magnetorquer-only systems have become increasingly viable for cost-constrained missions because of their simplicity and low power consumption [4].

Two primary types of magnetorquer designs exist: Air-core magnetorquers and ferromagnetic-core magnetorquers. Air-core magnetorquers consist of a coil of wire, typically copper, without a ferromagnetic core within. These designs prioritize low weight and compact form factor because they eliminate the requirement for a heavy and bulky ferromagnetic core. However, due to their lack of a ferromagnetic core, they provide weaker magnetic dipoles, and thus weaker torques, than equivalent ferromagnetic-core magnetometers. Most air-core magnetorquers are wide and flat to minimize their space usage, and some are embedded within PCBs, allowing them to have a minimal form factor. Ferromagnetic-core magnetorquers consist of a coil of wire wrapped around a ferromagnetic core. The ferromagnetic core allows them to produce stronger dipole moments per unit mass and power but adds hysteresis and additional mass to the system. In addition, they must be in the traditional solenoid-style coil design in order to hold their ferromagnetic cores, and thus cannot be flat or embedded in PCBs, increasing their space usage. Villalba-Alumbreros et al. optimized various core geometries using FEA simulation and validated with Hall effect sensors, providing a roadmap for custom designs [5].

Other actuators used for attitude control include reaction wheels and thrusters. Reaction wheels produce torque by spinning a flywheel contained within the satellite, causing an equal and opposite torque on the satellite body.

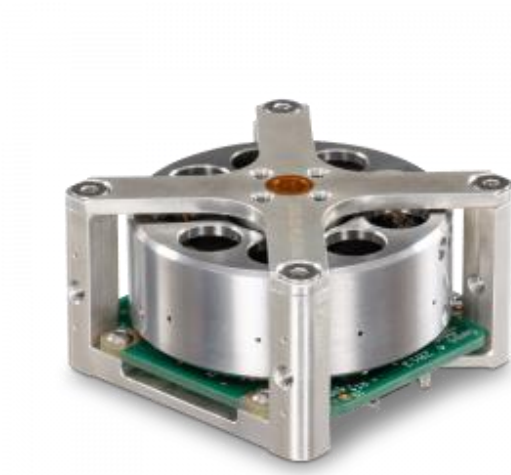


Figure 2: Photo of a commercial-off-the-shelf reaction wheel [6]

Reaction wheels, as seen in Figure 2, can produce thousands of times more torques than magnetorquers, but they have a fundamental limitation: saturation. Once a reaction wheel reaches its maximum speed in one direction, it becomes saturated and cannot produce additional torque in that direction until it is slowed down, or desaturated, using external torques from magnetorquers or thrusters. In addition, reaction wheels are complex systems and are prone to failure, with 18-20% of all spacecraft failures from 2007-2022 being linked to reaction wheel subsystem failures [7].

Finally, thrusters can produce torques comparable to reaction wheels and can also produce linear forces, allowing for a satellite's orbit to be changed after launch. However, they are extremely complex systems as at least 6 thrusters are necessary to control all three degrees of freedom because thrusters can only push on the spacecraft, not pull. They also have a limited lifetime due to their need for propellant and require propellant storage units. Because of this, the use of thrusters is beyond the scope of this project [8].

Detumbling

When a spacecraft is released from its launch vehicle, it launches with some large initial angular velocity with no preferred axis due to the unpredictability of the launch and release process. This rotation could be in the tens of degrees per second, typically dependent on the launch provider, and must be reduced for the spacecraft to power the solar panels, communicate with ground stations, determine its attitude, and perform the scientific experiment. Detumbling is the process by which this is achieved, and the B-dot algorithm, developed in 1976, remains the universal standard for initial spacecraft detumbling operations. This algorithm applies control torques according to $\mathbf{m} = -k\dot{\mathbf{B}}$, where $\mathbf{m}$ is the magnetic dipole moment, k is a gain constant, and $\dot{\mathbf{B}}$ is the rate of change of the magnetic field measurements [9]. This reduces rotational energy whenever the magnetic field rate of change and angular velocity vectors are not parallel, causing the spacecraft to attain a constant attitude relative to the magnetic field, eliminating tumbling.

Attitude Pointing

Pointing accuracy is defined as the error between the target pointing attitude and the current pointing attitude. CUSLI has a requirement for $< 8^\circ$ pointing accuracy. This pointing

accuracy is required to achieve 99% payload accuracy and solar panel efficiency because both the solar energy received by both the payload and the solar panels vary with the cosine of the angle between their normal vectors and the sun vector, and $\cos(8^\circ) \approx 0.99$. To achieve this accuracy, several magnetorquer-only approaches exist:

Lyapunov Control: Lyapunov control takes advantage of the fact that controllability changes as the satellite orbits and promises eventual convergence to a desired state. However, this is not active pointing control and does not allow for quick changes in pointing targets but can maintain a fixed orientation like for coarse sun-pointing. The expected pointing accuracy reported is $\pm 5\text{-}8^\circ$ (68% of time) and up to $\pm 13^\circ$ (worst case) with a 2-5 hour acquisition time after deployment [10].

PD vs. LQR Controllers: Torczynski, et al compared PD and LQR (Linear Quadratic Regulator) controllers for magnetorquer-only satellite control systems and found that PD controllers outperform LQR controllers for settling time and steady-state error, though LQR is slightly more energy-efficient in terms of actuator energy. One key challenge with LQR controllers is that they guarantee stability only for time-invariant systems, not periodic systems like satellites in orbit, where the Earth's magnetic field changes periodically throughout the orbit. In order to apply LQR controllers to this problem, Torczynski, et al performed Floquet analysis, a mathematical technique for verifying stability in periodically time-varying systems, for stability verification. They then also did this analysis on PD controllers to verify their stability. They found that for polar orbits such as the one they used, at least $\frac{1}{4}$ orbit is required to guarantee settling from arbitrary disturbances due to the Earth's magnetic field rotating throughout the orbit [11].

Related Work

Custom Magnetorquer Design and Fabrication

Li et al. from York University documented the complete design cycle for a CubeSat attitude control system, including custom magnetorquer development. Their rod-type magnetorquers featured a 5.7 mm diameter permalloy core with 6,063 wire turns, achieving a maximum magnetic dipole of 0.37 Am^2 with a mass of 28 g per magnetorquer. They developed custom coil-winding machinery using stepper motors controlled by an ATmega644P microcontroller to achieve precise, automated winding. Their work demonstrated that university teams could successfully design and manufacture magnetorquers in-house, though their focus on ferromagnetic-core designs introduced complexity through hysteresis effects and core geometry optimization [12]. Villalba-Alumbreros et al., on the other hand, took a more analytical approach, using finite element analysis (FEA) simulation to optimize ferromagnetic core geometries, and Hall effect sensors to validate their designs [5]. Although both works focus on ferromagnetic-core designs and our project differs by pursuing an air-core magnetorquer design, they inform our project by demonstrating that designing and fabricating custom magnetorquers in-house is feasible.

Project Description

Functional Requirements

The functional requirements of the system have two main portions- the hardware requirements and the software requirements. The software portion of the system must determine the required actuation effort for the system given its current state and its desired state. The hardware portion of the system must then perform the desired actuation to reach the desired state, consuming power in the process. A diagram of the functional decomposition is shown below in Figure 3.

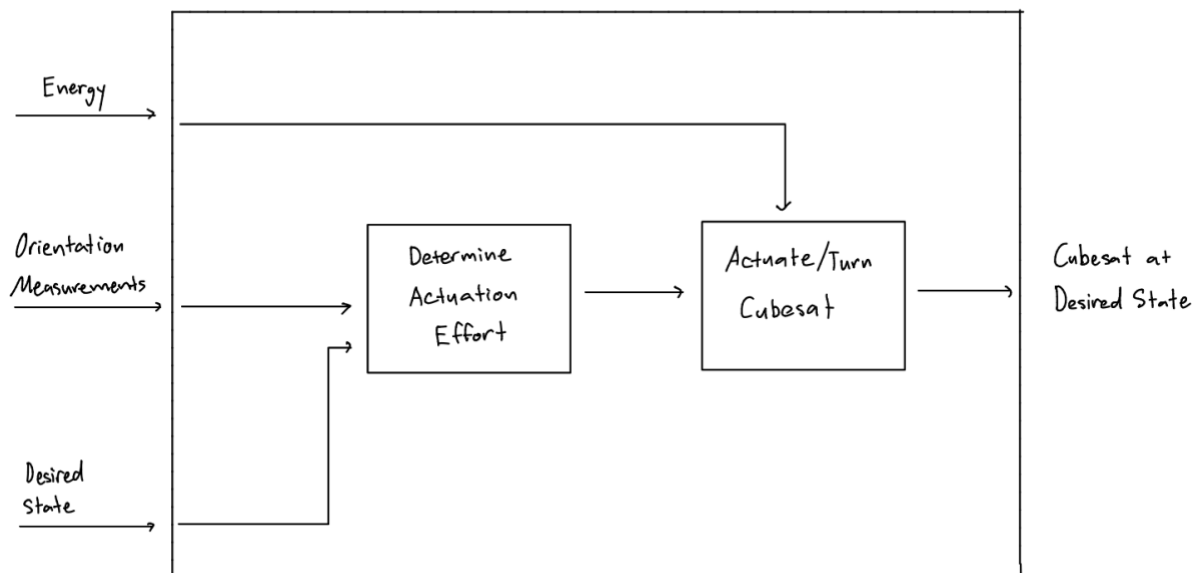


Figure 3: Functional Decomposition

The physical requirements for the system fall on the hardware portion. These are listed in Table 1.

Table 1: Physical constraints

Constraint:	Requirement:
Size	$\leq 10 \times 10 \times 3 \text{ cm}$
Weight	$\leq 200 \text{ g}$
Magnetic dipole strength	$\geq 0.06 \text{ Am}^2$
Energy consumed for detumbling	$\leq 44.4 \text{ Wh}$

The size and weight constraints come from discussion with groups responsible for the other CubeSat subsystems. The space and mass remaining after those systems provide the constraints for the space and mass allowable for the attitude control subsystem.

The maximum power consumption comes from the maximum power capacity of the CubeSat's batteries. This is because the most power-intensive part of the CubeSat's mission is detumbling, where the CubeSat must reduce a large initial angular velocity to zero without being able to recharge. Thus, the main function requirement for the attitude control system is to be able to detumble the satellite without running out of power. This requirement, however, is meaningless without a timeframe, because a magnetorquer's calculated power consumption must be multiplied by a length of time to find its total energy consumption. Therefore, a time for detumbling needs to be found before the energy consumption restraint can be fully fixed.

The minimum dipole strength constraint comes from comparison to prior work on similar systems. Torczynski et al. designed a system to detumble a similar CubeSat in 2 hours using three magnetorquers with dipole strength 0.06 Am^2 [11]. Using this as the minimum

constraint for magnetic dipole strength guarantees that this satellite will be able to detumble in a similar timeframe.

Another possible constraint which could arise on satellite attitude control is the ability to track a moving target. This could be a significant constraint, as the torques required to actively track a fast-moving target could be far greater than those required to point in a single static direction. In addition, tracking might require torques in specific directions which magnetorquers cannot produce due to the orientation of the Earth's magnetic field. We therefore analyzed various cases to determine if the attitude control system would need to be able to track moving targets.

For the experimental payload, precise tracking is not required as the sun is extremely far away, so its motion is negligible on the time scales at which the satellite operates. For ground station communication, on the other hand, tracking may be required for adequate data transfer because the satellite moves quickly relative to the Earth. Therefore, we analyzed the specific parameters of our expected orbit to see if this would be required. For a typical LEO orbit and an on-orbit antenna half-power beamwidth of 40 degrees, the CubeSat has ~76 seconds of connection as it passes over the ground station with no active tracking. With a conservative downlink rate of 100 bps, this gives ~7.6 kb of data downlinked per orbit, which was determined to be enough for our experimental payload. Therefore, tracking is not required.

To verify the feasibility of meeting these constraints simultaneously, we looked in literature to find the efficiencies of comparable magnetorquers. In a survey of possible magnetorquer designs, Villalba-Alumbreros found a worst possible efficiency of $0.267 \text{ Am}^2/\text{W}$

[5]. Thus, if the 3 magnetometers for this system each have a similar efficiency, and must produce 0.06 Am^2 in order to detumble in 2 hours, they will consume a total of

$$0.06 \text{ Am}^2 / (0.267 \text{ Am}^2 / \text{W}) * 3 \text{ coils} * 2 \text{ hours} = 1.34 \text{ Wh}$$

This is well within the constraint of $< 44.4 \text{ Wh}$ consumed for detumbling, showing that this set of constraints is feasible to fulfill simultaneously.

The only constraint on the software portion of the system is that it must lead to a final pointing error of $< 8^\circ$. This was done because both the solar panels and the scientific payload of the CubeSat depend on incident sunlight, and the intensity of incident sunlight on a face is proportional to the cosine of the angle between the face normal and the direction of the incident light. To achieve $> 99\%$ accuracy for these subsystems, the angle between the face normal and the incident light must be less than 8° , leading to this constraint.

Final Design

Given our technical constraints and the previous analysis, we decided that it was feasible to build the attitude control system using solely magnetorquers, rather than other actuators. Magnetorquers have the advantage of being lighter, cheaper, simpler, and less failure prone than other actuators, as mentioned previously, so we decided to design a system using only magnetorquer to take advantage of these qualities.

To fit the attitude control system in the given space constraint of $10 \times 10 \times 3 \text{ cm}$, a configuration which consists of 2 long magnetorquers and one wide magnetorquer was chosen. This was inspired by products such as the ISISPACE iMTQ board, as seen in Figure 4, which uses

this configuration to fit 3 magnetorquers into a small rectangular space such as that available here [13].



Figure 4: Image of ISISPACE iMTQ board [13].

The final design will include an optimized three-axis magnetorquer system with their currents controlled by H-bridge/motor controllers and a microcontroller running the control algorithm. For verification, a one-axis magnetorquer demo will be made by allowing the magnetorquer to rotate about one axis with minimal friction and detumbling and pointing tests will be performed. Once the one axis hardware-in-the-loop testing is complete, a three-axis control algorithm will be developed and tested. Due to the difficulties of achieving and measuring frictionless three-axis motion, a simulation environment will be developed using MATLAB's Simulink to test the three-axis control algorithm for detumbling and pointing.

Other Constraints

According to the research conducted so far, there are not relevant codes or standards for CubeSats or small satellites. This makes sense, as the existence of small satellites is a quite new phenomenon, and thus industry best practices have not yet been established.

When the CubeSat is ready for launch, the launch provider will likely have some standards or requirements for launch. However, because a launch provider has not yet been chosen, it is not possible to know these criteria yet. Given that the design of this attitude control system is similar to those found in prior literature, it can be assumed that these requirements will not necessitate any major changes to the design of this subsystem.

Design Criteria

In discussions with representatives of other CubeSat subsystems, it was concluded that the overall CubeSat mission would not benefit by the attitude control system reducing its mass, volume, or power consumption because none of the other subsystems are constrained by mass, volume, or power consumption. Thus, in determining an optimal magnetorquer design, it was decided to use a score function which consists of dipole moment divided by mass. This is because dipole moment is proportional to torque (because $\vec{\tau} = \vec{m} \times \vec{B}$) and torque is related to angular acceleration through $\vec{\tau} = I\vec{\alpha}$, where I is the moment of inertia. Moment of inertia is related to mass, although not linearly, so angular acceleration will increase with higher torque and decrease with higher mass. Thus, optimizing for dipole moment per unit mass effectively optimizes for higher angular acceleration. In addition, in order to encourage the score function to prioritize stronger magnetorquers over weaker magnetorquers, an additional 0.1 times the

magnetic dipole was added. Thus, the final score function was $\frac{|\vec{m}|}{m} + 0.1|\vec{m}|$. This is useful for the satellite because a higher angular acceleration leads to faster turning speed, which allows less time to be wasted in turning and more time to be spent performing mission functions such as taking experimental readings, recharging batteries, and communicating with Earth.

Ethical implications

The most critical ethical issue faced during this project is the responsibility of sending an object into low earth orbit, contributing to the growing amount of space debris. As per the ASME code of ethics, mechanical engineers are held ethically responsible to consider environmental impact and sustainable development in the performance of their professional duties [14]. By creating a low-cost attitude control system, our project potentially contributes to the growing quantity of debris in low-earth orbit by making CubeSats and low-earth orbit more accessible. In space, orbital debris is the primary threat to spacecraft and can damage spacecrafts in the best-case scenario and cause catastrophic failures in the worst. Introduced by Donald Kessler, the Kessler Syndrome, also known as collisional cascading, is the runaway chain reaction of collisions caused by debris, which, in turn, creates more debris and more collisions until the orbit is rendered no longer usable [15]. This scenario is theoretical, however it would have extremely negative effects for the humanity as a whole should it come to pass, so this system's role in contributing to it must be taken seriously. However, CUSLI's CubeSat is also 1U, the smallest standard form factor at 10x10x10 cm, which pales in comparison to the other pieces of debris in orbit. While the launch of the CubeSat cannot be prevented, the value of CUSLI's CubeSat can be maximized by ensuring the attitude control system is reliable and keeps the satellite functional during mission fulfillment rather than letting it become debris prematurely, which is the most ethical approach to the issue within the project's constraints.

Status

So far this semester, the technical constraints and basic design were established. Once this was done, specific magnetorquer designs were chosen through an optimization process. To do this, 5 variables were considered for an overall system design: number of coils in long magnetorquer, diameter of long magnetorquer, number of coils in flat magnetorquer, diameter of flat magnetorquer, and thickness of wire. Then, a cost function was chosen, as mentioned above, and applied to a generic search space of possible magnetorquers, as shown in Figure 5.

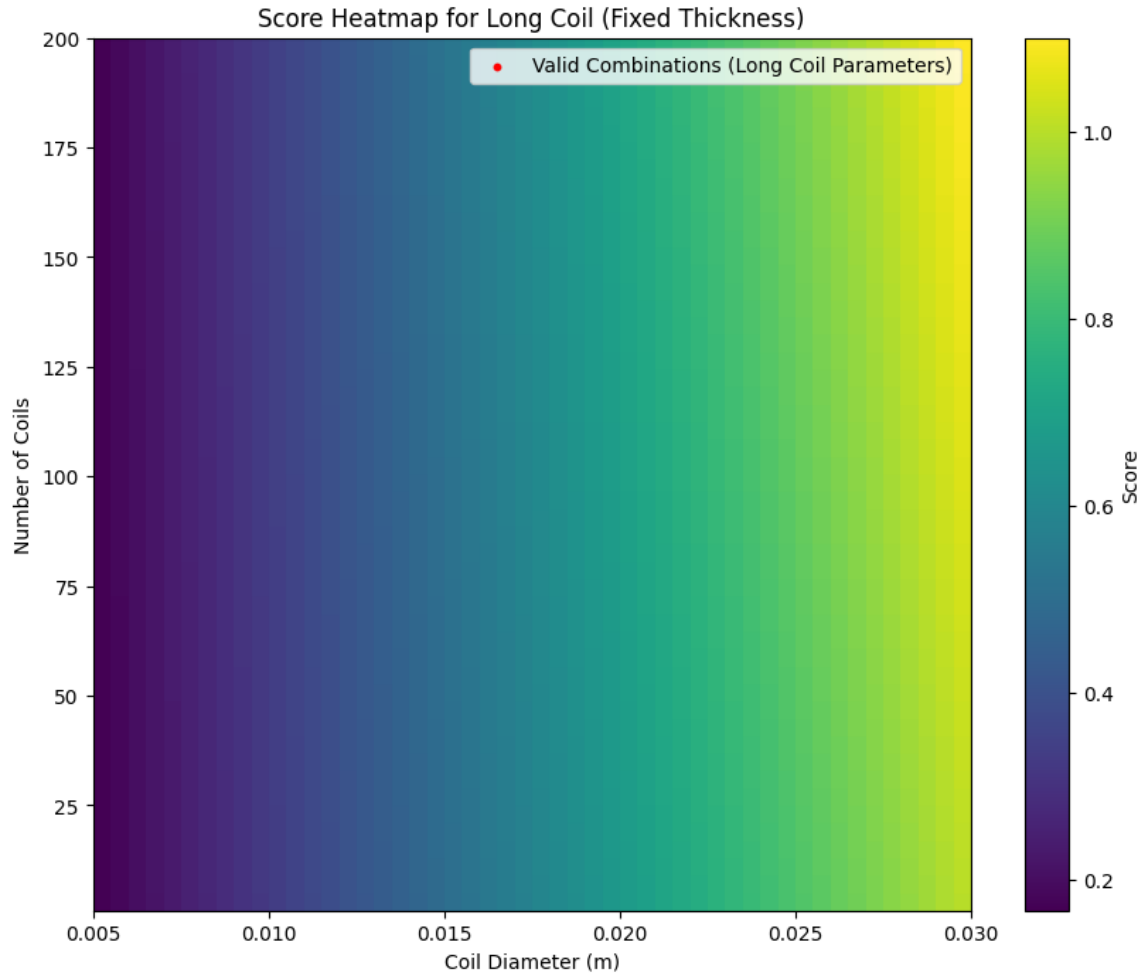


Figure 5: Heatmap of score function

Then, the search space was filtered to only include combinations of magnetorquers which meet the system technical requirements listed in Table 1. This yields an optimal design, which meets the technical requirements while maximizing the score function. The parameters for this design can be seen in Table 2.

Table 2: Optimal Magnetorquer Parameters

Parameter	Value
Number of coils in long	105

Diameter of long coils (cm)	3
Number of coils in flat	47
Diameter of flat coils (cm)	9
Wire thickness (mm)	0.25

After this theoretical optimization, physical verification was performed in order to verify the calculations of magnetorquer properties. To do this, a random magnetorquer was fabricated with 84 coils, a 3 cm diameter, and 0.8 mm wire. These values were chosen because of the materials available at the time. This coil is shown in Figure 6: Image of fabricated coil in testing. The coil also contains a cavity within which holds an AA battery to allow the coil to be powered without adding torque from connections to an external power supply.

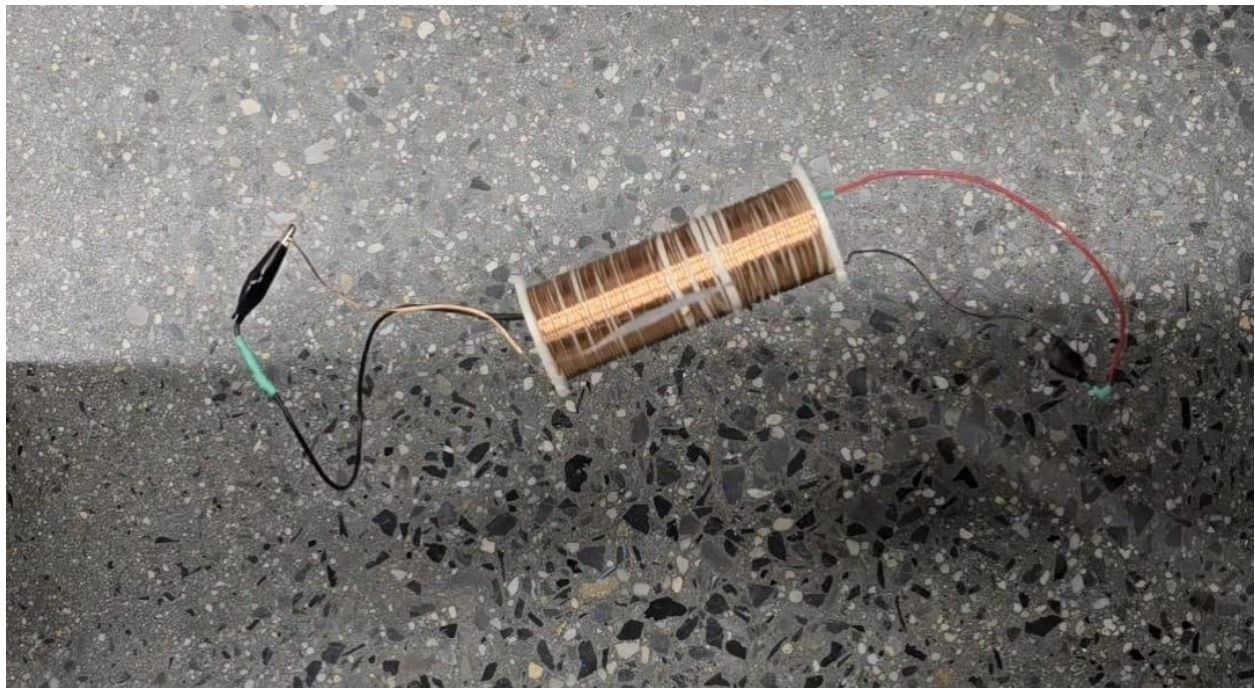


Figure 6: Image of fabricated coil in testing.

Once this coil was fabricated, it was tested in three ways. First, its resistance was measured and compared with the calculated values to verify testing safety and the physical parameters chosen. For this specific coil, the anticipated resistance value was $0.26\ \Omega$ and the measured value was $0.3\ \Omega$, confirming the calculations. Then, a gaussmeter was used to measure the magnetic field produced by the coil and this value was compared with the theoretical value. For this coil, the anticipated field strength in the center of the coil was $1.01\ \text{mT}$, and the measured value was $1.0\ \text{mT}$, again confirming the calculations.

For both of these tests, the expected values were calculated using basic equations. The resistance of a wire can be calculated as $R = \frac{\rho L}{A}$, where ρ is the resistivity of the wire, L is the length of the wire, and A is the cross-sectional area of the wire. Expected magnetic field values can be calculated using the formula for field of a solenoid which is $B = \mu_0 \frac{NI}{l}$, where μ_0 is the magnetic permeability of free space, N is the number of coils in the solenoid, I is the current through the solenoid, and l is the length of the solenoid.

Finally, dynamic testing was performed by suspending the coil with an internal power source from a fishing line and allowing it to rotate freely without connecting the power source to the coil. Once steady state was attained, the power source was connected to the coil, and the coil was allowed to attain steady state again. This test was intended to estimate the torque provided by the coil; however, issues were encountered because the coil did not reach steady state easily, and repeatedly experienced disturbances from fluctuations in the air.

A subsequent dynamic test was performed by hanging the same coil in the anechoic chamber, which was intended to reduce these disturbances. This largely worked, and the coil

did not experience the same fluctuations as the previous test. However, the coils oscillations still did not subside over the duration of the test, making it hard to determine an exact steady state. Once the battery was connected, the coil began oscillating about a different steady state to the one without the battery, showing that the coil is indeed producing a torque, as predicted. However, it was not possible to characterize this torque because doing this would require knowledge of the torsional string constant of the fishing wire, as well as detailed knowledge of the Earth's magnetic field in the anechoic chamber, neither of which was available at the time. In addition, the battery likely began to die during this test, causing the current through the coil and the torque produced by the coil to slowly decrease over the course of the test, further complicating attempts to measure it.

Because of these difficulties with hanging-type tests, future testing will likely involve other methods to ensure freedom of movement. We are currently considering floating a device on a pool of water or using some sort of air bearing to levitate it. The main advantage these methods have over the tests performed previously is that they lack the spring force inherent to the fishing line- because the fishing line has a natural resting orientation, it will constantly cause a torque on the device, which will interfere with measuring the torque produced by the magnetorquer. Air or water levitation, on the other hand, will not cause any torque on the device. In addition, a water test would be quite heavily damped. This would provide an advantage by more quickly reducing any oscillations caused by external fluctuations but may make the torque effects of the magnetorquer itself harder to see.

In addition, regardless of the specific test method used, a better containment for the magnetorquer is necessary. This is because in previous tests the magnetorquer was powered

simply by an AA battery inside the casing. Because of the simplicity of this design, it was not possible at all to measure or control the current through the magnetorquer. Future designs for rigorous testing will require larger batteries and a voltage regulator to strictly control the current through the coil. In addition, a microcontroller will be included to allow for more complex tests, including 1-axis controller testing.

Next Steps

The anticipated next steps for this project are listed in the Gantt chart in Figure 7: Gantt chart of next steps.

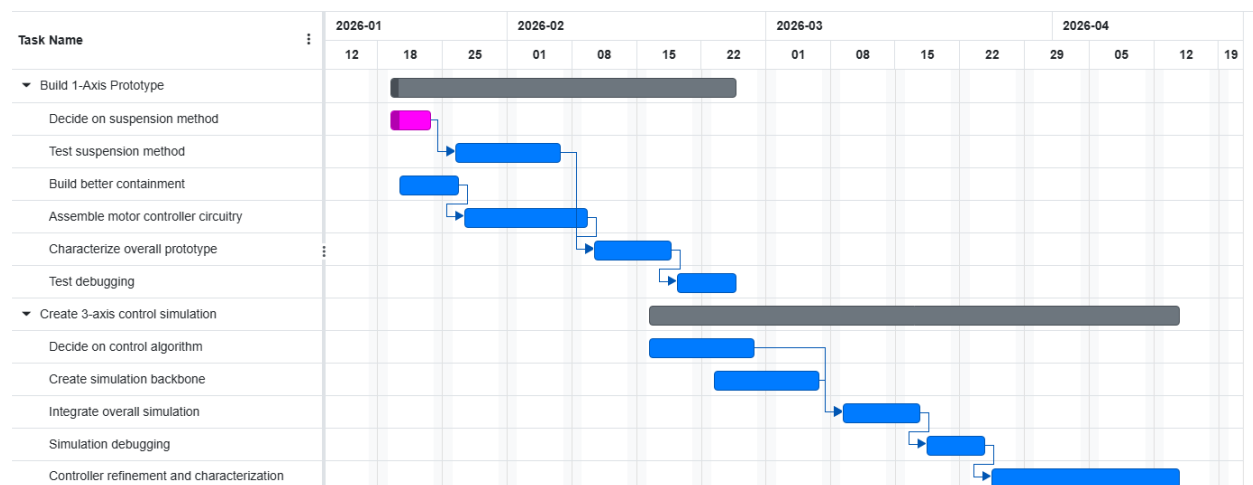


Figure 7: Gantt chart of next steps

The descriptions of each step are listed in Table 3.

Table 3: Description of next steps

Decide on suspension method	Decide on the method to be used to achieve 1-axis rotational freedom. Potential options are floating on water or using air suspension
Test suspension method	Test suspension method using current assembly to find any unexpected issues or disturbances

Build better containment	Build a containment which can hold the magnetorquer, batteries, motor controller circuitry, and microcontroller. This task is currently assigned to other CUSLI members
Assemble motor controller circuitry	Assemble circuit which uses motor controller and voltage regulator to draw power from batteries and power magnetorquers with consistent current. Will most likely be assembled on a breadboard
Characterize overall prototype	Determine torque produced by magnetorquer. We should have a fluctuation-free and consistent-current test bed at this point
Test debugging	The testing will most likely not go smoothly, so this week is to allow for any quick fixes necessary
Decide on control algorithm	Based on research, decide on what control algorithm to use for 3-axis control. B-dot will be used for detumbling, something else for pointing.
Create simulation backbone	Create models for 3DOF rotation, orbital motion, magnetorquers. Magnetorquers will depend on exact parameters learned from 1-axis testing
Integrate overall simulation	Integrated previously chosen controller with other simulation components to get a full simulation
Simulation debugging	Errors will likely arise somewhere in this process- this will give time to fix them
Controller refinement and characterization	Tune and characterize controller (detumbling time, pointing speed, steady-state error).

Note that this schedule ends in mid-April, whereas the project will be in progress until mid-May. This gap allows for flexibility in case some steps take longer than expected. If extra time is available after all steps have been completed, it may be possible to begin work on a 3-axis physical prototype. However, this will likely be more than one month's worth of work, so any progress there will likely be a stepping stone to other groups' future work.

Due to the small size of the main team working on this project, tasks are not being split between team members. Instead, both team members are working together on tasks, with weekly meetings to ensure consistent progress and additional meetings whenever necessary to accomplish specific goals.

In addition to the main team members of this project, there are some other students who are interested in joining CUSLI and helping with the attitude control system. These students have been given the task of designing the physical structure of the 1-axis demo, which will hold the magnetorquer, battery, and electronics in place. This group has only started working recently, so a consistent schedule has not yet been established, but it is anticipated that they will be meeting and coordinating with the main team weekly.

Conclusion

This project's goal was to design a low-cost attitude control system for CUSLI's CubeSat. We have shown that in-house fabricated magnetorquers alone are sufficient for this purpose without needing expensive commercial off-the-shelf solutions such as reaction wheels and thrusters. We have begun developing a magnetorquer-based system that fits all our constraints: it takes up less than 10x10x3 cm, weighs under 200 g, uses less than 44.4 Wh during detumbling, and can point accurately enough ($<8^\circ$) to keep the payload and solar panels working at over 99% efficiency.

We optimized the magnetorquer design and chose a configuration with two long magnetorquers (105 coils, 3 cm diameter) and one wide magnetorquer (47 coils, 9 cm diameter) using 0.25 mm wire. To make sure our calculations were right, we built and tested a prototype. The results matched our predictions closely, which confirmed the validity of our theoretical calculations and allows us to further proceed with building optimal magnetorquers.

Next semester, we'll improve our test setup to better measure the torques the magnetorquers produce, develop a three-axis control algorithm, and develop a simulation environment to test the three-axis control algorithm. This project does more than just give CUSLI a working attitude control system at a fraction of the usual cost, it makes space more accessible by showing that it is possible for university teams to design, build, and test their own magnetorquers from scratch.

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